

popquiz Free Fall

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:28pm

Quiz Instructions

Free-fall means no air resistance. $a = -10\text{m/s/s}$ about



Question 1 20 pts

A freely-falling watermelon falls with constant

☐

acceleration.

☐

velocity.

☐

speed.

☐

distances each successive second.



Question 2 20 pts

In class we discussed the experiments done by Galileo using inclined plane. He came to the conclusion that the total distance covered by a sliding object after a time t is proportional to

☐

initial velocity only

☐

time elapsed squared

☐

time elapsed cubed

☐

time elapsed



Question 3 20 pts

In class we discussed the experiments done by Galilei with inclined planes. To measure the time, he

☐

used his pulse

☐

used a pendulum

☐

used a time watch



used a bell



Question 4 20 pts

An object falls freely from rest on a planet where the acceleration due to gravity is 20 m/s^2 . After 5 seconds, the object will have a speed of



100 m/s.



20 m/s.



50 m/s.



5 m/s.



10 m/s.



Question 5 20 pts

The position $x(t)$ of a particle as a function of time t is given by the equation

$x(t) = (3.5 \text{ m/s})t - (5.0 \text{ m/s}^2)t^2$. What is the average velocity of the particle between $t = 0.30 \text{ s}$ and $t = 0.40 \text{ s}$?

- steps: 1) From the kinematics equation, find the initial velocity V_0 and the acceleration (negative)
- 2) Draw a number line that shows $[X_0=0, V_0, t=0]$ and $[X_1, V_1, t=0.3\text{s}]$, and $[x_2, V_2, t=0.4\text{s}]$ so 3 points.
- 3) Find $V_1=V(0.3\text{s})$ and $V_2(0.4\text{s})$. Use one kinematics equation
- 4) Use the equation (average velocity) = $(V_1+V_2)/2$

note: you can also use displacement = x_2-x_1 then use average = displacement/time elapsed.

$x_1=x(0.3\text{s})$ and $x_2=x(0.4\text{s})$

tricky



4



-3



2

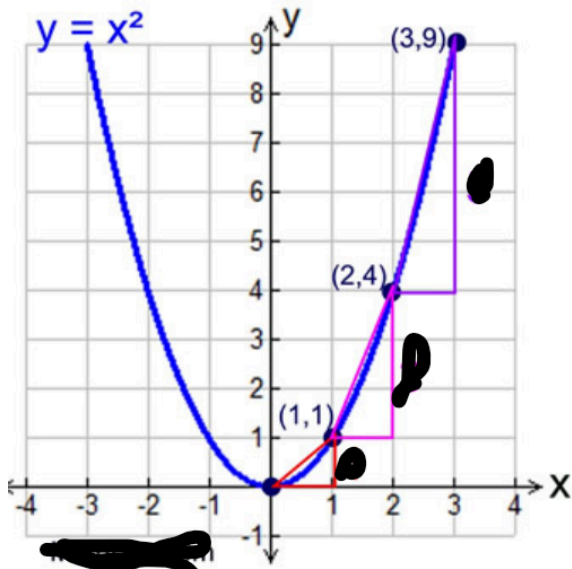
☐

0

⋮

Question 6 5 pts

bonus



Whats the slope of the red tangent line, the pink one and the purple one

☐

1, 2, 3

☐

0, 4, 9

☐

1,3,5

☐

1, 2, 4

⋮

Question 7 2 pts

If this is a graph of position vs time. What is the physical quantities that the slope represents

☐

displacement

☐

instantaneous velocity

☐

average acceleration



instantaneous acceleration

Not saved

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